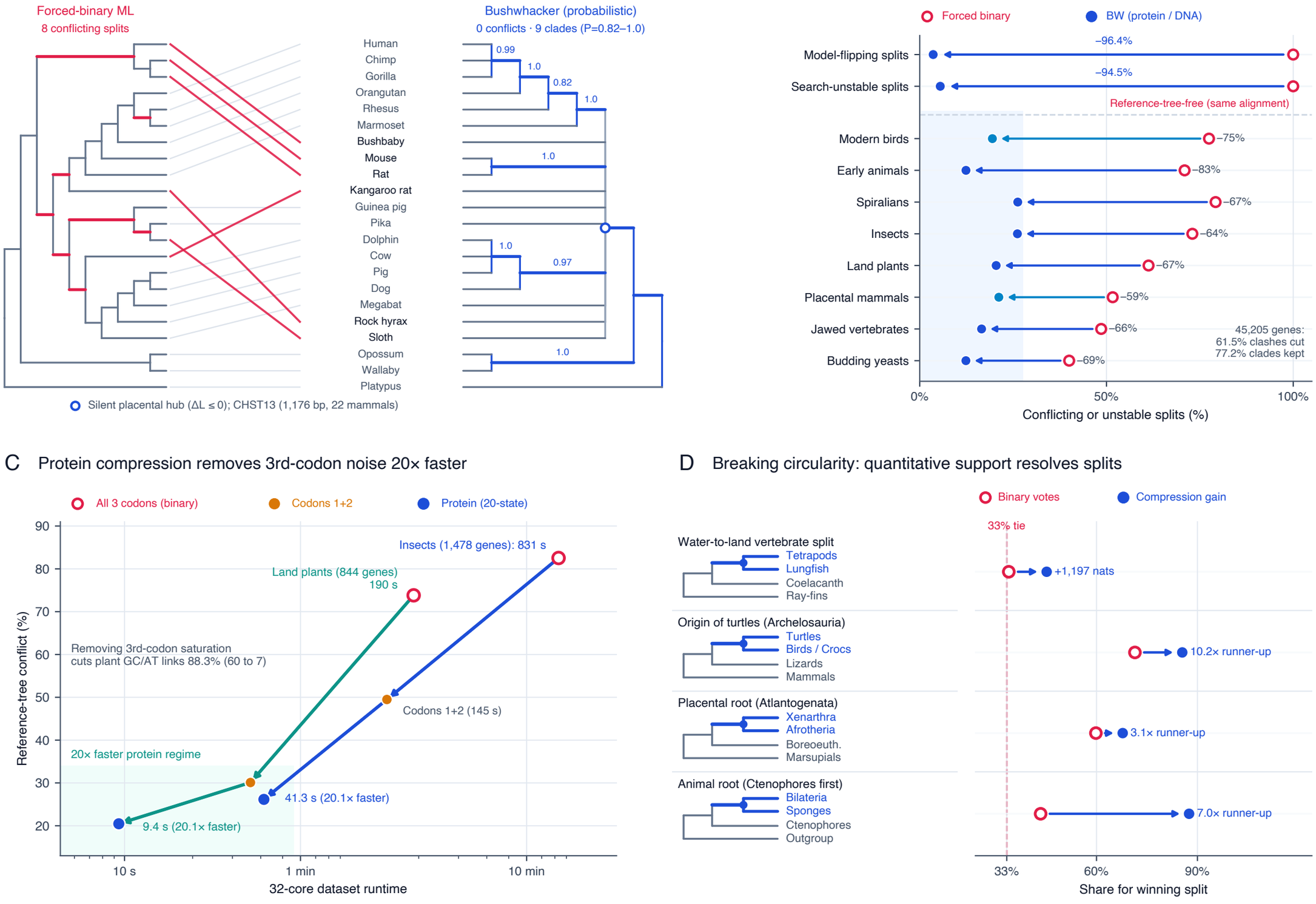
A
Untangling an empirical mammalian gene tree
Forced-binary ML
8 conflicting splits
Bushwhacker (probabilistic)
0 conflicts · 9 clades (P=0.82–1.0)
Human
Chimp
Gorilla
Orangutan
Rhesus
Marmoset
Bushbaby
Mouse
Rat
Kangaroo rat
Guinea pig
Pika
Dolphin
Cow
Pig
Dog
Megabat
Rock hyrax
Sloth
Opossum
Wallaby
Platypus
0.99
1.0
0.82
1.0
1.0
1.0
0.97
1.0
Silent placental hub (ΔL ≤ 0); CHST13 (1,176 bp, 22 mammals)
B
Same-alignment flips and conflict across 19 radiations
Forced binary
BW (protein / DNA)
Model-flipping splits
Search-unstable splits
−96.4%
−94.5%
Reference-tree-free (same alignment)
Modern birds
Early animals
Spiralians
Insects
Land plants
Placental mammals
Jawed vertebrates
Budding yeasts
−75%
−83%
−67%
−64%
−67%
−59%
−66%
−69%
45,205 genes:
61.5% clashes cut
77.2% clades kept
0%
50%
100%
Conflicting or unstable splits (%)
C
Protein compression removes 3rd-codon noise 20× faster
All 3 codons (binary)
Codons 1+2
Protein (20-state)
Reference-tree conflict (%)
90
80
70
60
50
40
30
20
Insects (1,478 genes): 831 s
Land plants (844 genes)
190 s
Removing 3rd-codon saturation
cuts plant GC/AT links 88.3% (60 to 7)
Codons 1+2 (145 s)
20× faster protein regime
41.3 s (20.1× faster)
9.4 s (20.1× faster)
10 s
1 min
10 min
32-core dataset runtime
D
Breaking circularity: quantitative support resolves splits
Water-to-land vertebrate split
Tetrapods
Lungfish
Coelacanth
Ray-fins
Origin of turtles (Archelosauria)
Turtles
Birds / Crocs
Lizards
Mammals
Placental root (Atlantogenata)
Xenarthra
Afrotheria
Boreoeuth.
Marsupials
Animal root (Ctenophores first)
Bilateria
Sponges
Ctenophores
Outgroup
Binary votes
Compression gain
33% tie
+1,197 nats
10.2× runner-up
3.1× runner-up
7.0× runner-up
33%
60%
90%
Share for winning split